

List of commands for LENSTOOL ([Kneib+96](#); [Jullo+10](#))

GENERAL USAGE

Manage interactions with DS9 frames:

DS9 > File > XPA > Connect/Disconnect

Plot object catalogs in a DS9 frame:

`./pelli.sh [.cat/.dat]`

Notes: If the LENSTOOL scripts are placed in a directory included in the `$PATH`, then one can call `pell_i [.cat/.dat]` directly.

Generate critical lines and caustics at the source redshift (z_s):

`best.par > cline > nplan 1  $z_s$`

Outputs: `ci.dat`, `ce.dat`

Smoothness parameters: `limitLow`, `limitHigh`

Plot in DS9: `./pcl.sh clean/noclean int/ext`

Make lensed image predictions:

`best.par > runmode > image 1 [.cat]`

Format of the input image catalog:

[ID] [RA] [DEC] [a''] [b''] [PA°] [z_s] [mag]

For a point image, one can set the `a` and `b` radii to 0.01.

Outputs: `image.all`, `source.dat`

Export image coordinates from DS9:

DS9 > Regions > Save Selected Regions > XML

Generate time-delay maps (τ in years):

`best.par > runmode >`

`time 1 [imsize in pix]  $z_s$  [output.name.fits]`

Generate magnification (μ) maps:

`best.par > runmode >`

`ampli 1 [imsize in pix]  $z_s$  [output.name.fits]`

`ampli 2 # absolute values (ignoring parity sign)`

One can open these maps in DS9 to generate isocontours of magnification or arrival time-delay:

DS9 > Analysis > Contours Parameters >

Generate > Apply > File > Save

Save current DS9 frames with all parameters:

DS9 > File > Backup (then to open it use [Restore](#))

Generate a `best.par` file:

`./bayesBest.sh [.par]`

Compute the probability density functions of the lens model free parameters using the default MCMC sampler:

`./bayesMap.sh [.par]`

Transfer data from a cluster to the current directory:

`rsync -avP [username]@[server]:/path/to/data .`

RAY-TRACED SP RECONSTRUCTION

Introduction to ray-traced source-plane reconstruction using custom-made PYLENSTOOL routines, following the source reconstruction methods implemented in LENSTOOL.

`python do_master_source_reconstruction.py`

`-m [best.par] -r [.reg] -i [.fits] - $z_s$  [int]`

`--sampling_image [int] # echant`

`--sampling_source [int] # s_echant`

`--lenstool_name /path/to/directory/bin/lenstool`

Regarding the IP and SP sub-sampling factors, one may use $\text{echant} \equiv \langle \mu \rangle$ and $\text{s_echant} \equiv \frac{1}{2} \text{echant}$ as initial values.

Apply a magnification correction to the data:

`cleanlens > flux 1`

`flux 0`: no μ correction, used for source rotation velocity and velocity dispersion (e.g., ALMA line moments 1 and 2).

Output FITS images:

- Reconstructed source plane at z_s

`[...]_reconstruction.fits`

- Reconstructed image plane at z_{obs}

`[...]_lensed_reconstruction.fits`

Plot PSFs or synthesized beams in the source plane:

`best.par > runmode > image 1 [psf_ellipses.cat]`

Format of the input catalog, which contains a list of circular or elliptical PSFs/beams located across the lensed image:

[ID] [RA] [DEC] [a''] [b''] [PA°] [z_s] [mag]

Output: `source.dat`

Display the results in DS9:

`open [...]_reconstruction.fits,`

`then ./pelli.sh source.dat`

One can convert PSF/beam sizes from arcsec to parsec scales using the [arc2parc](#) tool or [cosmology calculators](#).

USEFUL LINKS

- CRAL Git repository of LENSTOOL
<https://git-cral.univ-lyon1.fr/lenstool>
- CRAL Git repository of PYLENSTOOL
<https://git-cral.univ-lyon1.fr/CALENDS/pylenstool>
Notes: Contact Guillaume Mahler to use the latest routines.
- Strong Lensing Cluster Atlas Database hosted at the Laboratoire d'Astrophysique de Marseille (LAM)
<https://data.lam.fr/sl-cluster-atlas/home>
- List of DS9 command line options
<https://ds9.si.edu/doc/ref/command.html>